

Abhishri Ajit Medewar

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EDUCATION

Master of Computer Science , Arizona State University	August 2021 - May 2023
Relevant Coursework: Machine Learning, Deep Learning, Cloud Computing	<i>GPA 3.8</i>
Bachelor of Computer Engineering , Pune University	August 2014 - May 2018
Relevant Coursework: Data Structures, Database Management Systems, Software Engineering	<i>GPA 3.7</i>

SKILLS

Programming Languages: Python, C++, C

Frameworks and Libraries: PyTorch, Tensorflow, Scikit-learn, Numpy, Scipy, Pandas, Matplotlib, Seaborn, Keras, OpenCV, BeautifulSoup, AWS, Boto3, Docker, S3, Hadoop, Kubernetes, MLflow, Weights & Biases, ONNX, TensorRT

Web Technology and Database: Flask, FastAPI, React, Angular, HTML, CSS, MySQL, PostgreSQL, MongoDB

Operating System/Developer Tools: Linux, Windows, Git, Jira, VS Code, Sublime text, Visual Studio, PyCharm

EXPERIENCE

Machine Learning Engineer, vivoVerse, Austin, USA October 2023 - Present

- Developed and deployed an object tracking model, achieving a 2% accuracy improvement. Converted the **model into ONNX format** to achieve faster inference speed and ensure smooth deployment in production environment.
- Implemented and integrated **data pipelines into ML platform** to facilitate model training processes. Utilized **MLflow** for comprehensive testing, version control, and real-time monitoring of machine learning models.
- Implemented a vision **transformer** from scratch for classification of images with dimensions of 5000x3000, leading to a **5% enhancement** in classification accuracy.
- Developed streamlined procedures for the storage, retrieval, and manipulation of large scale (terabytes) datasets utilizing **PostgreSQL**, enhancing overall efficiency.

Machine Learning Engineer Intern, Intel Corporation, Santa Clara, USA June 2022 - April 2023

- Implemented **distributed training** and hyperparameter optimization for a video frame interpolation model, resulting in a 40% reduction in inference time and significant performance enhancement.
- Reproduced cutting-edge research paper results by leveraging **Docker** for efficient and streamlined training.

Machine Learning Engineer, Tata Elxsi, Pune, India July 2018 - June 2021

- Implemented State-of-the-Art region-based object detector using **Faster-RCNN** with **98%** accuracy and average inference time of 3sec/image on **Jetson TX2**, with image dimension of 2592x1944.
- Deployed an InceptionV2 and Faster-RCNN based object detector on **Nvidia Jetson TX2** using Tensorflow C++ API and bazel.
- Implemented **K-means clustering algorithm** for detecting small dust particles on camera lens, anti-reflection glass for optical system with overall accuracy of **95%**.

PROJECTS

U-Net-Based Image Background Segmentation and Editing [[Github](#)] Summer 2023

- Implemented U-Net architecture from scratch, achieving an impressive 98.90% accuracy (dice score) in background segmentation using the Human Segmentation Dataset.
- Employed TensorRT to convert the trained model into an optimized TensorRT format, resulting in a significant 6x performance boost.

Digit Recognition- Deployment with Kubernetes and Docker [[Github](#)] Spring 2023

- Developed an ML application for handwritten digits classification, using Flask server and Docker for seamless deployment.
- Utilized Kubernetes clusters on AWS EKS and GCP GKE for streamlined deployment and performance monitoring.

DeepLearn: Simplified CNN Training and Evaluation for Any Dataset [[Github](#)] Fall 2022

- Implemented State-of-the-Art CNN architectures from scratch such as GoogLeNet / InceptionV1, AlexNet, VGG, LeNet, and ResNet with custom dataloader.
- Utilized **FastAPI** to develop RESTful APIs, enabling streamlined model deployment and seamless integration within application.

Deep Learning Approach for Human Stress Detection[[Github](#)] Summer 2018

- Developed custom CNN for detection of stress level (0-9) by using FER2013 dataset to identify the facial emotion of the person from an image and correlating the emotion percentage with stress level with 85% accuracy.
- Utilized Flask to develop APIs for serving models, facilitating seamless deployment of model.

ACHIEVEMENTS

- Received **Tata Elxsi Bravo Award** for outstanding work in machine learning model fine-tuning.
- Exhibited strong leadership by driving Proof of Concept work at Tata Elxsi and securing funding from key clients.